Municipality of Tuy

COMPREHENSIVE LAND USE PLAN 2023-2033

Volume I: The Comprehensive Land Use Plan

MESSAGE OF THE MAYOR

The formulation and the implementation of the Comprehensive Land Use Plan of the Local Government Unit Tuy will serve as a tool in Identifying the gaps embedded in the utilization of land for all stakeholders of the Municipality. It will enable several bureaucratic institutions to propel growth and development with creation of a mechanism which aims not only to supply relevant information on the resources persistent in Tuy but would also eradicate the inimical environment which stipulates a disparity on access to basic fundamental services offered to every claim holder of the Municipality. Furthermore, the implementation of the Comprehensive Land Use Plan Into effect our just and relevant reiteration on sustainable development which, thru the plan can Be fully realized as the instrument equips, both public and private entities with appropriate facts and data on the kind of development which espouses the resources abundant the Municipality without setting aside environmental dynamics. On the other hand, I would like to extend my warmest felicitations of the staff of the Office of the Municipal Planning and Development Office for tirelessly working on every facet of the comprehensive plan which is an evident manifestation of their personnel's' dedication and reckoning in ensuring the unfathomable fulfillment of a mission specially to ease all forms of detriments in a community thru bureaucratic initiatives and innovative platforms. May this plan serve as a genuine reflection of my mantra of harnessing progress from the grassroots level of Our society as displayed in my battle cry "Asenso ng Bayan SiguRADO".

JOSE JECERELL C. CERRADO
Municipal Mayor

Asenso ng Bayan SiguRADO

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MESSAGE OF THE VICE-MAYOR

The CLUP serves as a roadmap guiding our development initiatives, ensuring that growth is managed in a sustainable and equitable manner. It is a reflection of our collective vision for the future of our community, encompassing various aspects such as land use, infrastructure, and environmental conservation.

In recent years, our dedicated team has been hard at work conducting assessments, analyzing data, and engaging with stakeholders to shape the foundation of our CLUP. Through these collaborative efforts, we have identified key opportunities and challenges that will influence the course of our municipality's development.

I am pleased to report that significant progress has been made in drafting the updated CLUP document tailored specifically to the needs and aspirations of Tuy. This document integrates innovative strategies aimed at fostering balanced growth, preserving our natural heritage, and enhancing the overall quality of life for our residents.

However, our work is far from complete. We recognize the importance of your involvement in this process, as your insights and perspectives are invaluable in shaping the future of our municipality. Therefore, I encourage each of you to actively participate in upcoming public consultations and forums where you will have the opportunity to review the draft CLUP and share your feedback.

Together, let us continue to work hand in hand to build a brighter and more sustainable future for Tuy—one that reflects our shared values and aspirations.

Thank you for your unwavering support and dedication to the progress of our municipality. Your commitment to our community is truly commendable, and I am confident that by working together, we can achieve great things for Tuy.

Nawa'y magpatuloy ang Maaasahang Responsableng pag-Asenso sa bayan ng Tuy!

ARMANDO P. AFABLE
Municipal Vice Mayor

ACKNOWLEDGEMENT

To God be all the glory!

The Local Government of Tuy, Batangas wishes to honor and gratefully appreciate the invaluable support, contributions and hard work of the following:

- Our Municipal Mayor Jose Jecerell C. Cerrado, Vice Mayor Armando P. Afable and Honorable Members of the Sangguniang Bayan.

- The Executive Offices being spearheaded by the Planning and Development Office through EnP Narciso V. Mulingbayan (MPDO) and all members of the CLUP Technical Working Group, as well other agencies of which in many ways have contributed also for this endeavor.

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FOREWORD

The Municipality of TUY, Batangas being a third class municipality envisions the future with ardor to promote sustainable development and equity of its resources amongst its constituents; with this in mind, what better plan it could have but through its Comprehensive Land Use Plan (CLUP). Being one of the mandated documents each LGU should make and implement, Tuy prepared hopeful for this feat to finally be enacted for its approval and implementation. This will lay the foundation and pathway for achieving development and growth through a rationalization of allocation and proper use of land resources; Tuy being a land-locked municipality. It will also be the basis for the zoning laws that on the other hand will restrict certain developments to protect the environment, help promote social engagement and enhance community involvement in the conservation of its land and provide consideration for a vision of a robust economy, industry and transportation progress. It will project the private and public land uses in accordance with the future spatial organization of economic and social activities keeping in hindsight proactive measures for sustainable environment and disaster-resilient communities.

Ultimately, it will serve as reference for local leaders in formulating development policies and identifying programs and projects for inclusive growth promotion. As Professor Ernesto M. Serote (Author of Rationalized Local Planning System in the Philippines) quoted: "effective land use planning is possible only when ideology, politics and techniques converge". Thus, tackling all the processes and requirements for its completion, having the skills, knowledge and resource of technical persons and officials behind, it will not only provide a path for protection, conservation and sustainable use of land resources of the municipality but also offer an appropriate and ideal framework that will best meet the needs of the present inhabitants while safeguarding the land resources for future generations as well.

As such, it is but the fervent goal of the leaders and stakeholders of this municipality to adhere and undertake the systematic land use planning through CLUP because the efficient and responsible management of land resources is expected all the time and a great responsibility of the implementers for better governance and exemplary public service.

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BRIEF PROFILE

The Municipality of Tuy is geographically located at 14° 1'9.56"N latitude and 120°43'48.65"E longitude and is found at the northern western portion of Batangas City. It is bounded on the north by the Municipality of Nasugbu, on the southwest by the Municipality of Lian, on the south by the Municipality of Balayan and Calaca City, and northeast by Tagaytay City. Out of the 22 barangays, four are categorized as urban barangays while 18 are classified as rural. The urban barangays include barangays Luna, Burgos, Rizal, and Rillo which are also located in the Población area of the municipality.

There is only one identified rock in Tuy namely quaternary-pliocene rocks. This falls under volcanic rocks which are formed by the cooling of molten magma on the earth's surface. According to the Bureau of Soils and Water Management, there are three (3) classifications of parent soil material in the Municipality of Tuy. Majority of the land area has eutrandepts with eutropepts parent soil material having 6638.786 has. This comprises Barangays Toong, Mataywanac, Magahis, San Jose, Bolboc, Rizal, Luna, Acle, Dao, Lumbangan, Rillo, Tuyon Tuyon, Putol, and Guinhawa. This type of parent soil occurs regularly on the steep erosional surfaces of basic volcanic formations. Based on the Bureau of Soils and Water Management Office, the Municipality of Tuy is generally characterized to have slope rise of 2 – 5% with fine loam type of soil with Oxic Us tropepts Parent Material comprising of 4,555.57 has or 47.71% of the total land area.

Based on the data of the US Geological Survey, the Municipality of Tuy has a topographical range characteristic of 10 to 720 meters above sea level (masl). Barangays Guinhawa, Dalima, and Sabang have the lowest elevation among the 22 barangays of the municipality with the range of 10 to 30 masl. Based on the 10-m interval of elevation of Tuy, the majority (54.18 percent) of the municipality are under the 100 meters elevation. This covers largely parts of barangays Sabang, Bayudbud, Dalima, Luntal, Mataywanak, Talon, Palincaro, Malibu, Lumbangan, Rillo, Tuyointuyon, Putol, Guinhawa, Dao, Acle, Rizal, and Luna.

Tuy as an inland municipality has a sloping characteristic of Level to Rolling with sloping rise of zero (0) to 18%, and it constitutes 6,288.26 has or 65.86% of the total land area of the municipality. According to the survey made by the Community Environment and Natural Resources Office (CENRO), 82.82 percent of the whole municipality is classified as Alienable and Disposable (A&D). The remaining 16.30 percent is classified as forest land.

Based on the Land Cover of NAMRIA in 2015, most of Tuy's land cover was identified to be for annual crop with 70.61 percent or 6,742.20 hectares of the total land area. Some 1,342.81 hectares or 14.06 percent were classified as covered with perennial crops. Generally, Tuy is rich with sugarcane, cassava and corn plantations since the majority of its land area is for annual crop cultivation.

The climatic type of the Municipality of Tuy under the Corona classification falls under Climatic Type 1 defined by two pronounced seasons: dry from the months of November to April, and wet or rainy the rest of the year. This seasonal change affects the behavior of Taal lake as manifested by lake overturns which happens during wet season and algal blooms during dry season.

The Municipality of Tuy is exposed to various hydro-meteorological, geologic, and volcanic hazards due to its geographical location and characteristic. The most frequently-occurring disasters in the municipality are floods. Of the 22 barangays, 13 are exposed to flood, 5 to landslide, 1 to liquefaction, and all barangays are exposed to earthquake and ashfall. Based on the data provided by the Mines and Geosciences Bureau, the 96.40 hectares of the Municipality of Tuy is highly susceptible to flooding while 108.17 hectares are moderately susceptible. Barangay Guinhawa has the largest area susceptible to flooding at 87.32 hectares. Overall, only 204.58 hectares or 2.14 percent of the municipality is susceptible to flooding. Majority of the municipality is at low susceptibility to landslides at 6,353 hectares. There are 1,231 hectares at moderate susceptibility, while 1,687 hectares at high susceptibility. Liquefaction takes place when loosely packed, water-logged sediments at or near the ground surface lose their strength in response to strong ground shaking. Overall, 2,843 hectares are not exposed to liquefaction, 1,919 hectares have low exceedance, 4,315 hectares are at moderate exceedance, and 386 hectares are at high exceedance.

The 2022 existing land use of the Municipality of Tuy was derived using satellite imagery from Google Earth 2022. It was validated through field visits, ground truthing activities, and land use workshops. Based on this, it can be inferred that the general land use of the municipality is largely agriculture. The agricultural area comprises about 71.79 percent of the municipality's total area or about 6,854.80 hectares. These are mostly covered by sugarcane, cassava, coconut, mango, dragon fruit, upland rice, lowland vegetables, corn, banana, coffee, cacao, peanut, and other fruit trees and root crops. Bulk of commercial areas are located in barangay Rillo, Luna, Rizal, and Burgos. Commercial areas account for a total area of 13.42 hectares (0.14 percent of the total land area). Agri-industrial areas are scattered in different barangays and Industries are also an emerging economy in the municipality. Dwelling units of the residents of the municipality fall under the Residential area. This accounts to a total area of 269.73 hectares (2.83 percent). Most residential areas are in the Poblacion barangays and along the road network since these are accessible to basic services and road networks.

The municipality posted a total population of 43, 734 for the censal year 2015 and 46, 519 on censal year 2020. About 16.02% or 7, 009 people are residing in urban barangays, while 39, 510 people are located in the rural barangays. Barangay Luntal is the most populated with 3, 624 people that corresponds to 7.79 percent, followed by Magahis with 3, 525 people. On the other hand, Barangay Burgos is the least populated with 625 people or 1.34 percent of the municipality's total. There are a total of 11,458 households in the municipality in 2020. Households in the urban barangays totals to 1,710 while those in the rural areas totals to 9,740 households. Thickly populated rural barangays are Magahis, Luntal, Putol, Sabang and Bolboc. Some of these barangays are near the Poblacion and area along the national road. The average household size in the Municipality is at 4.06 which implies that members of the household are mostly parents with two (2) children.

The 2020 Census of Population indicates that the school-going population is 17, 574 which covers 37.84% of the total household population. Among them, elementary students have the highest population (5,487) followed by high school students (5, 382), pre-school students (3, 468), and

college students (3, 237) which is the least populous. Looking into the productive population, it comprises 64.11% (29, 771) of the total, 35.89% of which is dependent on the working group. Among the dependent population, 79.86% or 13,312 are young and 20.14% or 3, 357 are senior citizens. Consequently, the working group should be actively employed to support the dependent population. The pyramid below shows a bigger male population compared to the female population in Tuy for the last two censal years. In 2020, there is a 1.63 percent edge from male over female. The sex ratio is 1.02 meaning that there were 102 males for every 100 females in 2020. Dependent population is dominated by the young with very little contribution from the old. In 2020, 35.89 percent of the household population belongs to dependents (i.e., young and old), with more females than males. From the total dependent household population, 79.86 percent are child dependents, while the rest are old dependents. There are generally more young male dependents (29.59 percent) than female dependents (27.73 percent).

Housing statistics in 2015 revealed a total of 10,073 housing units; of which 96.37% are occupied and the rest are vacant. From the previous censal year 2010, the number of households increased by 8.50%. In 2015, the ratio of household population to occupied housing units was four (4) which means that the average household consists of the parents and two (2) children. The Tuy Rural Health Unit is located in Barangay Luna (Poblacion) occupying a total area of 160.62 square meters. The manpower includes one (1) municipal health officer, one (1) public health nurse, five (5) rural health midwives, one (1) rural sanitary inspector, one (1) medical technologist, one (1) radiologic technician and one (1) visiting dentist. It is further augmented by barangay health workers (BHW), barangay nutrition scholars (BNS) nurses and midwives who form the NDP (Nurse Deployment Program) and RHMPP (Rural Health Midwife Placement Program of the National Government. Elementary education is well provided in the Municipality of Tuy. There are 18 public elementary schools which are found in the different barangays of the Municipality except Barangay Acle which has no elementary school because of its proximity to Agoncillo Elementary School at Barangay Tuyon-tuyon. The central school is Gregorio Paradero Elementary School which is located in the urban area and accommodates all elementary pupils from the Poblacion barangays namely: Burgos, Luna, Rillo, and Rizal.

The Tuy Police Station is situated in the Poblacion along Gomez Street fronting the Tuy Town Plaza which occupies a total area of 280 sq. m. The peace and order situation is being safeguarded by the Philippine National Police and supported by the volunteer tanod brigades in the barangay. Furthermore, there are two (2) police outposts located at Barangay Rillo and Tuy Public Market; both outposts are assisted by barangay tanods for immediate response during emergency situations. The stations have barangay tanods and volunteers for crime and drug watch, they are also tasked with maintaining peace and order. The Tuy Fire Station is also located in the same area. It occupies a total area of 50 sq. m. Tuy has a town plaza with a total land area of 8,133 sq. m. which is located in the Poblacion in front of the Municipal Building. The town plaza has a covered court, open basketball court and a children's playground. A municipal fitness gym is also located within the Municipal Plaza. Sports facilities are also available in different public and private schools. Social welfare facilities in the Municipality include a daycare center, senior citizen's building, gender and development building, and LGU drop-in centers. These facilities are used in the provision of day care services, issuance of senior citizen's ID and purchase booklets, senior citizen activity area, venue for occasional activities such as therapy center for ASD Children, and serves as temporary shelter for different sectors such as children, persons with disability and senior citizens. In 2022, the Municipality has 24 day care centers with 23 child development workers and two (2) teacher aides. There is no day care center in Barangays Rillo and Burgos.

The Municipality of Tuy has a total Agricultural area of 3,829.67. A total of 1,304 MT contributed to the annual volume of production of various crops for the year 2022. The largest area is from sugarcane crop which recorded 3,442.17 hectares. This is followed by cassava with a total of 960 MT harvested from 80 hectares of agricultural land. Lastly, the harvested total of 130 MT of corn from 50 hectares of land of the municipality. The commercial areas in Tuy are mostly in Poblacion Area. There are boutique shops, dry goods and commercial buildings. Meanwhile, the public market is located in Brgy. Rillo having an area of 6,118 sqm. In Brgy. Luntal, there are food chains for the locals to go to.

According to the United Nations World Tourism Organization, tourism is defined as the activities of persons traveling to and staying in places outside their usual environment for not more than one consecutive year for leisure, business and other purposes. The municipal government of Tuy has been investing in the tourism sector as it capitalizes the resources it currently possesses, which is in congruence with the objective of the Department of Tourism to promote the agri-tourism sector in every community. A total of five (5) tourist establishments in the Municipality are listed in Table 4-14. Among them, two (2) are resorts, two (2) are outdoor activities (trekking and biking), and an art exhibit. All are accessible through land transportation.

The Municipality of Tuy has a total road network of about 164.68 kms. The 13.74 kms or 8.35% of the total road network are national roads. These roads include the road section from Brgy. Guinhawa to Putol, Tuyon-tuyon, Rizal, Luntal, Mataywanac, and Sabang. Provincial road 1 is located in the stretch in Brgy. Rillo, Luna, and Bolboc, while provincial road 2 covers roads in Brgy. Luna, Mataywanac and Toong with a total length of 15.65 kms or 9.51 percent. The municipal road has a total length of 0.74 kms or 0.45% situated at Basa. St to Bolbok; and the barangay road has a total length of 134.54 kms or 81.70%. A total of 101.66 kms kilometers or 61.73 percent of the road network are made of concrete while 48.39 kilometers or 29 percent are earth. Only 14.63 kilometers or 8.88 percent are asphalt. Power is necessary for the development of a locality. The Municipality of Tuy is being served by the Batangas I Electric Cooperative, Inc. which is located in Calaca, Batangas. In 2023, a total of 12,480 households is listed as concessionaires of the aforesaid electric cooperative. Households at Barangays of Luna, Rillo, Rizal and Tuyon-Tuyon are already energized, though these are not reflected on the table due to boundary issues.

The level of water supply service in the Municipality is Level III. This is a waterwork system which includes a source, a reservoir, a piped distribution network, and individual household taps. Domestic types of consumers are served by Tuy Water District and Barangay Water and Sanitation Associations (BAWASA). The Tuy Water District has 3,953 active service connections while BAWASA has 5,481 number of connections. Majority of the domestic consumers are supplied by Tuy Water District with an average consumption of 20.08 cubic meters. Brgy. Bayudbud has no water supply connection. Tuy have access to the information and communication technology through the internet/ mobile phone providers namely Globe, Smart, PLDT, Sun and Dito. Remote areas of the Municipality lack internet accessibility. Municipality of Tuy has only 13 cellular sites across 22 barangays. Students residing in remote areas have limited access to information and communication technology which are vital to new modes of learning. Investors must be encouraged for the installation of additional cell site networks in the locality.

I. NATIONAL, REGIONAL, AND PROVINCIAL FRAMEWORK

A. Sustainable Development Goals

After the Millennium Development Goals (MDGs), the participating countries of the United Nations (UN) once again affirmed their commitment towards ending all forms of poverty, fighting inequalities, increasing country's productive capacity, increasing social inclusion, curbing climate change, and protecting the environment while ensuring that no one is left behind over the next fifteen years. A new global plan of action was adopted by the UN member states in September 2015 entitled "Transforming Our World: The 2030 Agenda for Sustainable Development." With the aim to stimulate people-centered and planet-sensitive change, Sustainable Development Goals (SDGs) otherwise known as Global Goals is composed of 17 goals and 169 targets and 247 global indicators which are integrated and indivisible, global in nature, universally applicable, taking into account different national realities, capacities and levels of development, and respecting national policies and priorities. The 17 goals are listed below.

Goal 1:	End poverty in all its forms everywhere
Goal 2:	End hunger, achieve food security and improved nutrition and promote sustainable agriculture
Goal 3:	Ensure healthy lives and promote well-being for all at all ages
Goal 4:	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
Goal 5:	Achieve gender equality and empower all women and girls
Goal 6:	Ensure availability and sustainable management of water and sanitation for all
Goal 7:	Ensure access to affordable, reliable, sustainable, and modern energy for all
Goal 8:	Promote sustained, inclusive, and sustainable economic growth, full and productive employment and decent work for all.
Goal 9:	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
Goal 10:	Reduce inequality within and among countries
Goal 11:	Make cities and human settlements inclusive, safe, resilient, and sustainable
Goal 12:	Ensure sustainable consumption and production patterns
Goal 13:	Take urgent action to combat climate change and its impacts
Goal 14:	Conserve and sustainably use the oceans, seas and marine resources for sustainable development
Goal 15:	Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
Goal 16:	Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable, and inclusive institutions at all levels
Goal 17:	Strengthen the means of implementation and revitalize the global partnership for sustainable development

B. Philippine Development Plan

The Philippine Development Plan 2017-2022 is the first medium-term plan to be anchored on the 0–10-point Socioeconomic Agenda and is geared towards the Ambisyon Natin 2040, which articulates the Filipino people's collective vision of a Matatag, Maginhawa, at Panatag na buhay para sa lahat.

AmBisyon Natin 2040 represents the collective long-term vision and aspirations of the Filipino people for themselves and for the country. It describes the kind of life that people want to live, and how the country will be by 2040. It is a set of life goals and goals for the country which defines the strategies to achieve the goals. It is like a destination that answers the question "Where do we want to be?". A plan describes the way to get to the destination. AmBisyon Natin 2040 is the vision that guides the future and is the anchor of the country's plans

All sectors of society, whether public or private, should direct their efforts towards creating opportunities for Filipinos to enjoy a Matatag, Maginhawa at Panatag na buhay. Government, in particular, must use its tools of fiscal, monetary and regulatory policies to steer the development path towards enabling Filipinos to attain their AmBisyon. This pertains to all dimensions of development: economic, human and physical capital, institutional, social and cultural.

AmBisyon aims that by 2040, the Philippines is a prosperous middle class society where no one is poor. People live long and healthy lives and are smart and innovative. The country is a high-trust society where families thrive in vibrant, culturally diverse, and resilient communities.

President Rodrigo Duterte issued Executive Order 27 in June 2017, that directs "all departments, offices, and instrumentalities of the national government" to undertake efforts to contribute to the full implementation of the PDP 2017-2022. The document is divided into three major strategic outcomes or pillars (1) Malasakit: Enhancing the social fabric, (2) Pagbabago: Inequality-reducing transformation, and (3) Patuloy na Pag-unlad: Increasing growth potential.

Malasakit: Enhancing the social fabric

The first pillar focuses on Filipinos who are proud of their nationality and trust their government. This aims to build greater trust between the people and the government. Public institutions aim to be cleaner, more efficient, effective, and people-centered through reduced corruption, seamless service delay, better administrative governance engaged and empowered citizenry and strengthened civil service. Moreover, fair and swift improvement in the justice sector is aimed. This is by enhanced justice systems and improved sector efficiency and accountability. Lastly, Filipino culture and values are targeted to be promoted greatly by valuing diverse cultures, infusing values for common good, developing creative excellence and strengthening culture-sensitive governance and development.

Pagbabago: Inequality-reducing transformation

Primarily, the second pillar aims for inclusive participation from marginalized sectors which can be achieved through different strategies. First, economic opportunities in the Agriculture, Forestry, and Fisheries (AFF) sectors will be expanded. Strategies include improving AFF productivity, increasing AFF-based enterprises, increasing stakeholders' access to value chains such as markets, increasing access to innovative financing and technology, and protecting access of farmers and fisherfolk to land and water resources. Also, economic opportunities in industry and services are promoted through increasing local and foreign investments, enhancing competitiveness and resilience of industries, improving market access, ensuring consumer safety, improving access to production networks, and boosting the stakeholders' access to finance and productivity, efficiency, and resilience. The third is

the acceleration of human capital development through improving nutrition and health and ensuring that everyone has lifelong learning and employment opportunities. Fourth, because individuals are vulnerable to different situations, reducing their vulnerability when these shocks arise may greatly help them. This is done through mitigating risks faced especially by vulnerable groups, managing economic risks, dealing with natural hazards, addressing governance and political risks, and achieving universal social protection for the citizens. Lastly, safe and secure communities are targeted through expanding the people's access to well-planned settlements.

Patuloy na Pag-unlad: Increasing growth potential

Accelerated and sustained economic growth is one of the focuses of the current administration. One of the strategies is ensuring that the workforce is employed and productive by monitoring the mortality rate and fertility rates, investing in service deliveries, and increasing the saving rates of the Filipinos. Science, Technology, and Innovation are endorsed through increasing its utilization in different sectors, investing in concerned enterprises, enhancing the creative capacity, and strengthening the collaboration among involved people.

Furthermore, the three pillars are supported by the formulation of an enabling and supportive economic environment, competitive and efficient Philippine market, solid foundation of peace and security, balanced and strategic infrastructure development, and ecological integrity. All these targets and strategies are accompanied by a proper monitoring and evaluation mechanism. The National Economic and Development Authority (NEDA) Secretariat is monitoring headline indicators from the local level to the national level.

C. Regional Development Plan 2023-2028

The CALABARZON Regional Development Plan (RDP) is anchored to the country's long-term vision – AmBisyon Natin 2040 and the desires of the residents of CALABARZON stipulated in the Regional Physical Framework Plan (RPFP). For the next 25 years, the RDP will be occupied in realizing the national vision through four identified areas namely, (1) toward building a prosperous, predominantly middle-class society where no one is poor, (2) toward promoting a long and healthy life, (3) toward becoming smarter and more innovative, and (4) toward building a high-trust society. CALABARZON is composed of five provinces, namely: Cavite, Laguna, Batangas, Rizal, and Quezon. It is composed of 25 congressional districts, 19 cities, 123 municipalities, and 4,011 barangays.

By 2022, the region aims to improve its economic and social status. Regional economic growth rate is targeted to be within 5.2 to 7.3 percent during the 2017-2022 timeframe. Poverty incidence is aimed to be reduced to 5.5 percent and unemployment rate is maintained within 7 to 9 percent. It is also targeted to reduce underemployment to 15 percent. The mentioned targets are to be achieved with the help of the advantages and potentials the region has. Some of these include the sustained demand in international market for electronics and semiconductor products, housing projects demand due to population increase, demand for various ICT services, finance and trade activities, increase in same day and overnight tourists, high investment opportunities from Overseas Filipinos (OFs) remittances and returning OFs, growth in high value crop and livestock production, and other positive economic impacts due to high capital infrastructure projects.

Unfortunately, the region is faced with certain challenges that it has to overcome to achieve their targets. These challenges are the slow growth of manufacturing sector and transport sector, inefficient transport network and poor intermodal transport system, absence of international airport, poor ICT infrastructure and facilities for smaller ports, lack of Tourism Master Plans and weak generation of tourism data, conversion of agricultural lands, increasing number of poor populations, and pollution.

Major strategies in the RDP may be classified into eight categories namely Industry, Agriculture, Maximizing Demographic Dividend, Human Capital Development, Services, Technology, and Innovation, Building Resilient and Secure Communities, and Infrastructure Development. Major strategies for each category are listed below.

Industry

- Strategic positioning producers and suppliers in the global value chain
- Promote CALABARZON industry clustering and ecozones
- Strengthen assistance and interventions to Micro, Small and Medium Enterprises (MSMEs)

Agriculture

- Research and promotion of climatic adaptation practices and innovations
- Maximize available agricultural lands for agro- forestry and agro-tourism & production of High Value Commercial Crops (HVCC) and integrated farming system
- Increased agriculture, forestry and fishery-based enterprises

Maximizing Demographic Dividend

- Intensify Labor, Financial Markets, and Human Capital interventions such as improving the school curriculum and strengthening academe-industry linkages to reduce job-skills mismatch
- Provide labor, financial literacy education and career guidance for students
- Improve health, nutrition and well-being.

Human Capital Development

- Expand access to nutrition and health services through more health personnel and provision of half-way facilities for remote areas and support infrastructure
- Expand health insurance coverage and social protection
- Increase competitiveness of graduates through full implementation of the Philippine Qualification Framework and benchmarking with ASEAN and global qualifications
- Strengthen academe-industry linkages on facilities, teacher and student training, and employment facilitation.

Services

- Improve capacities of LGUs, investors, stakeholders and institutions on Information Technology-Business Process Management (IT-BPM).
- Convergence of Tourism Master Plans in prospective tourism areas
- Aggressive tourism product development, marketing and partnerships with private sector

Technology and Innovation

- Accelerate technology transfer, commercialization and utilization.
- Strengthen open collaboration among actors in the science and technology innovations (STI) ecosystems
- Enhance creative capacity for technology generation, acquisition and adoption with global standards.

Building Resilient and Secure Communities

- Strengthen convergence mechanism for poverty alleviation

- Strict implementation and increase the number of social safeguard programs for the Elderly, PWD, children, occupational safety and health.
- Conduct of land inventory, informal settler families mapping and research on effectiveness of Local Shelter Plans

Infrastructure Development

- Strengthen interregional connectivity
- Improve access to tourist destinations, transport facilities and remote areas
- Improve mobility and flow of traffic
- Promote renewable and indigenous energy development and utilization

D. Batangas Provincial Development and Physical Framework Plan 2017-2022 (PDPFP)

The direction of Tuy's development shall conform to higher development plans such as the Philippine Development Plan (PDP) 2017-2022, CALABARZON's Regional Development Plan (RDP) IV-A 2017-2022, and Batangas's Provincial Physical Framework Plan (PDPFP) Batangas 2014-2022. Its development agenda shall follow suit and form part of the strategic frameworks of the mentioned plans.

The Strategic Framework Vision of the country in its Philippine Development Plan (PDP) 2017-2022 states that "By the end of 2022, more Filipinos will be closer to achieving their AmBisyon to have a *"matatag, maginhawa at panatag na buhay".* It furthers that the Administration will lay down a solid foundation for more inclusive growth, a high-trust and resilient society, and a globally competitive knowledge economy.

Tuy is described herein as having population and settlements with low development density. However, Tuy is geographically categorized as an Agricultural Municipality. The Local Government Unit invests in agricultural advancement in terms of production and distribution.

II. LOCAL GENERAL DEVELOPMENT PERSPECTIVE

A. Vision

TUY will be a Municipality of empowered and gender-responsive people, that values equality and recognition of every citizen's right, with God-loving leaders striving to achieve and provide the highest degree of importance to a clean, protected, safe and peaceful environment with disaster-resilient communities, having a progressive and sustainable economy with access to quality social services.

Table 1. Vision Success Indicators

Descriptor	Success Indicator
Institutional	
God-loving	• Additional service-oriented programs • No cases filed against the leaders • Transparency and accountability: Presence of transparency documents/activities
Social	
Empowered	• Capable, competent, and well-equipped individuals • Improved enrollment participation rate • Improved classroom-student ratio conforming to DepEd Standards • Presence of Provincial Hospital and RHUs with complete facilities • Decreased malnutrition rate by 50% • Decreased number of patients with lifestyle-related diseases by 50% • Increased care at all life stages (decreased MMR, IMR, CDR) • Improved ratio of doctors, nurses, midwife, and hospital beds conforming to DOH Standards • Decreased number of Informal Settlers Families (ISFs) • Presence of Socialized Housing Site • Percentage of clients provided assistance (educational, burial, and medical) • Maintained low crime rate
Gender-responsive	• Presence of VAWC desk (Violence Against Women and Children) • Presence of ERPAT (Empowerment and Reaffirmation of Paternal Abilities) • Presence of approved annual Gender and Development (GAD) Planning and Budget • Increased number of sectoral groups assisted (women, ERPAT, PWD, senior citizen, LGBT, children, solo parent, youth)
Economic	
Progressive	• Improved and Expanded Public Market • Promote fiscal discipline • Local Enterprises • Updated Revenue Code • Formulated and Implemented Local Investment and Incentive Code

Descriptor	Success Indicator
	• Increased employed people • Increased goat and livestock production • Formulation of Tourism Plan and Tourism Code • Establishment of the Eco-Bike Trail and Hiking Activities • Establishment of activities of painting of local artists to boost the tourism sector • Promotion of OTOP Biscocho, Bibingka and Sugarcane Leaves Handicrafts • Kambingan Festival
Sustainable	• Maintain high value crop yield • Maintain sugarcane crop production • Increase in poultry production • Dispersal of livestock animals • Distribution of vegetables seedlings • Availability of agri credit or loan program
Environment	
Clean	• Access to safe water supply • Improved management of poultry standardized requirements on the facility • Proper waste disposal • Presence of improved and expanded sanitary landfill • Garbage is regularly collected from 4 barangays and other business areas • Issuance of Environmental Clearance and Animal Inspection Certificate • Presence of Municipal Livestock and Poultry Task Force • Conformance of RA 9003 • Presence of Wastewater Treatment Facility
Protected	• Regular evaluation and monitoring of quarrying sites > presence of ordinance regulating quarrying activities • Presence of reforestation/afforestation activities • Continuous activities on tree-planting activities
Safe	• Street lights to all barangays • Decrease incidence of road accidents • Road signages
Peaceful	• Decrease in crime rate
Disaster-resilient	• Established signages in flood and landslide- prone areas

Descriptor	Success Indicator
	• Access to good network for proper communication • Additional equipment or tools for rescue in the MDRRMO
Infrastructure	
	• Percentage of paved roads increased • Number of municipal roads widened with complete features increased • Percentage of barangay roads in good condition increased • Number of permanent bridges increased • Number of ancillary road facilities increased • Number of facilities of public transport terminals upgraded and improved • Number of public schools with internet access increased • Proportion of households with electricity maintained

Source: CLUP Workshops

B. Development Potentials and Constraints

Various development potentials and constraints were identified to provide a quick outlook of the municipality. It will more or less provide a general picture of the municipality just to have a clear glance of the existing situation for cohesiveness in the planning perspective.

Development Potentials

- **Large area of cash/industrial crop and continuous effort for agricultural development** – The municipality is still largely agricultural in use, with sugarcane, cassava, and corn as the primary crops produced. Main sources of income are still reliant on the agriculture sector. The municipality also supports the sector by various programs and initiatives like seedling distribution, assistance and others. It also has a high number of agricultural-related activities and/or occupations.
- **Presence of National Grid Corporation of the Philippines (NGCP) sub-station** – The presence of NGCP facility makes the generation of energy from alternative sources feasible. To date, solar farms exist in Lumbangan (Project 1), Luntal, Bayudbud and Sabang (Project 2) and Bolboc (Project 3) by Citicore Solar Batangas with one occupying a total land area of 300 hectares more or less. This indicates Tuy's good spots on solar power investment.
- **Generally safe from hazards** – The municipality's exposure to hazards is quite manageable compared with the rest of LGUs in the province. Only 204.58 hectares or 2.14 percent is exposed to flooding. While 97.98 percent is exposed to landslide 67.13 percent of which is identified to have a susceptibility of low to none. When it comes to volcanic hazards, Tuy is not exposed to base surge. During the Habagat season, occurrences of ashfall and smog does not affect the municipality. In fact, during the Taal Volcano eruption, the locality offers evacuation areas for nearby municipalities affected by the eruption.
- **Relatively safe municipality** – There is a low crime rate stated in the municipality for both adults and children below 18 years of age.

- **Transition from local center to provincial center** – There is a proposed Provincial Hospital in Tuy that is expected to be completed by year 2025 which will cater most of the first District's populace. It is also expected to instigate growth and opportunities for the municipality in terms of more diverse services and structural development to offer and as well job opportunities.
- **Developing Adventure and Food Tourism** – Activity-based strategy is one way to boost the tourism sector of the municipality. Man-made bike trails traversing uneven terrains near mountains is the start of the municipality's efforts to flourish in the recreational aspect given that most of its resources are geared towards agricultural production and environmental attractions are few. At the same time, the municipality is making its way to establish its own identity through the "*Kambingan Festival*" and its fusion with the culinary industry.
- **Infrastructural Linkages** – The possible construction of Nasugbu-Balayan Expressway (NBEX) and East-West Road can pave the way to more accessible routes to the municipality and in the long-run can help expand the urbanized areas.
- **Rising industrialization opportunities –** The presence of big investors like Citicore Solar Batangas 1, Wilcon and other companies will increase job opportunities and income of the locality. On the other hand, there are also commercial farm contract growers like San Miguel Corporation, Bounty Fresh and CJoy.
- **Access to Potable Water Supply** – Almost 100 percent of the population in the municipality have access to treated water/potable drinking water being connected to the Water District/Level III Water System. It also has plenty of ground water sources that are tested to also be safe for drinking and consumption.
- **Ample areas suitable for future urban development** – Overlay analysis of land supply revealed that a total of 6,721.26 hectares or 71% percent of the total land area is suitable for future urban development.

Development Constraints

- **Dwindling sugar industry** – The Central Azucarera Don Pedro Inc. in Nasugbu, Batangas, considered as the main sugar milling center was closed. This is the major facility processing the bulk of sugarcane from Tuy. Inevitably, the initial impact is diminishing engagement in the sugarcane industry by the local farmers. Worst case scenario is converting sugarcane farms into non-agricultural purposes. The lack of investors in the sugarcane industry also contributes to the dwindling sugar industry.
- **Threat to air quality and sanitation** – The traditional poultry-raising has posed threat to health and air quality because the absence of modern technology to aid these situations cause nuisance and community complaints on pest infestation and air pollution.
- **Limited budget since Tuy is IRA dependent** – In 2020, out of the total budget of PhP182,161,018.91 million IRA constituted 135,126,231.96 or approximately 74 percent of the total Municipal budget. The 25.82 percent locally generated income accrued from local taxes, services and businesses, and other sources. Until today, the locality is still IRA dependent.
- **Limited natural assets to spur local tourism industry** – There are no coastal resources, hot springs, falls and other natural resources that are usually the main attractions for tourists. Only few nature-based resources for recreational/tourism purposes (e.g., falls, strip of Mt. Batulao) is present.
- **Aging local farmers and insufficient number of farm workers** – The inventory of land uses shows that 92.09 percent of the total municipal land area is considered agricultural found in

different barangays. However, it was observed that there are limited farm workers in Tuy. To augment the number of farm workers, "sakadas"or "magtatabas" are brought in from Quezon province to perform farm related activities during planting and harvest seasons.

- **Slow economic development** – Compared with nearby tourism-focused municipalities like Tagaytay and Nasugbu, and other adjacent LGUs, Tuy is considered backward in terms of economic development. In fact, in the 2022 DTI nationwide ranking of competitiveness index for economic dynamism indicators, Tagaytay City is 52nd, Lian is 167th, Nasugbu is 131st, Calaca City is 103rd, Balayan is 124th and Tuy is 238th . Commercial bank, which also indicates economic dynamism is not existing within the locality.
- **Inadequate Social Facilities** – Education facilities are not provided with complete support functional amenities like laboratories, shops, clinics, libraries, and others. The social welfare services and protective facilities are likewise considered insufficient.
- **Inadequate Parks and playgrounds** – Based on the existing standard parks and open spaces to population ratio, Tuy has insufficient area for such purposes. Because of its critical role to have a balanced psychosocial development, it is essential to nurturing a well-rounded community and therefore considered an important element of the overall development.
- **Insufficient Infrastructure and Utilities** – In 2022, approximately 6.12 percent of the total number of households in the municipality are not served with Level III potable water supply. The surface of 57.7 kilometers or 46 percent of the entire barangay roads is still earth and some of the drainage alongside of the roads caused flooding due to poor design.

C. Development Goal and Objectives

The development goal of the Municipality of Tuy is aimed at achieving inclusive growth and ensuring resilience and empowerment of its people. Specifically, the following objectives shall be pursued to fulfill the development goal of Tuy:

1. To diversify agricultural products, develop organic farming learning center, and engage in agritourism
2. To improve urban-rural linkages and connectivity with adjacent municipalities
3. To establish policies fostering sustainable development in different programs and projects initiated by both the public and private sector
4. To enhance the capabilities of all municipal officials and employees in all aspects of governance
5. To create a learning environment conducive for the development of practical and technical skills
6. To promote community participation in environmental protection and management
7. To support the promotion of local identity like the "Kambingan Festival", local artistry and other local culture and traditions
8. To protect forested areas and declare local watersheds in barangay Luntal, Sitio Palico.
9. To support the development and use of renewable energies

D. Development Thrusts and Strategies:

It is evidently shown that the municipality of Tuy remains as an **agricultural** town primarily supporting the sugarcane industry. Approximately 3,442.17 of agricultural areas are currently utilized for

sugarcane production where 1,273 farmers are economically dependent. However, it was observed that sugarcane is a dwindling industry in the municipality despite its predominance in agricultural production. From 2021 to 2022, sugarcane production areas decreased by 12.09 percent which may be attributed to the aging farmer population and limited farm workers. In the coming years, more sluggish economic performance of the sugarcane industry is expected especially with the closure of the Central Azucarera Don Pedro Inc. in Nasugbu, Batangas. However, Tuy would like to look at it in a more positive perspective. The LGU suspected that the dwindling sugarcane industry caused the emerging crop diversification practices among some of the local farmers. Hence, agriculture in Tuy is now transitioning from specializing in sugarcane production into crop diversification practices. This is a better strategy to lower the risk in production with the potential impact of climate change.

In promoting **agricultural development**, it is important to note that while shifting cultivation is already on the way, sugarcane production still plays a significant role in the local economy. It is still justifiable to **support the sugarcane industry** while adopting **crop diversification** at the same time. The agricultural development prospects intend to engage more local farmers in crop diversification to keep agricultural production profitable. In the development constraints, it was observed that the neighboring LGUs of Tuy, who have been engaged in the tourism industry, like Lian, have a more dynamic local economy. Tuy is looking forward to enhancing the local economy through **tourism development**. Despite limited natural tourism attractions in the locality, **agricultural-based tourism** products may be developed. Upgrading agriculture into agritourism is also expected not just to add economic value to the agriculture sector but to promote agriculture as a way of life, a local culture, and a noble job. To motivate crop diversification and agritourism, the municipality will focus on the following:

1. **Increase production per unit** through sustainable crop production practices such as crop rotation, multi-cropping, integrated farming, companion cropping, and organic farming; introduction of new high-yielding varieties of vegetable, fruits, and sugar seeds; and capability enhancement and technology transfer to farm families/local farmers;
2. **Improve agricultural value chains** by providing direct linkage from farm to market place or farm to table approach, and establishment of postharvest facilities; and
3. **Create a culture of identity and pride**. Tuy used to celebrate the "Kambingan Festival" to show identity and pride of having goat raising as one of the major economic Industries in town. Revival of such events and exploring other opportunities to build a culture of identity and pride will make agriculture more dignified.

While Tuy's development thrust and strategies are agriculture centered, the development portfolio for tourism will be further expanded by venturing into adventure tourism, ecotourism, and cultural tourism. Back in 2020, visitors enjoyed the biking trail in the municipality until the pandemic brought unfortunate circumstances. This time, the municipality would put its best foot forward to regain momentum in adventure tourism with the circumferential biking road as the flagship program. With regards to ecotourism, the municipality may not have a good number of natural assets to upstage the neighboring LGUs in the tourism business. However, Mt. Batulao is an outstanding asset that is already known to many mountain climbers. It has a breathtaking view of the Taal Volcano and a unique landscape. Likewise, ecotourism activities in local watersheds and riversides are also expected to foster protection of the area while generating economic gain at the same time. For cultural tourism, the LGU is looking into the revival of the "*Kambingan Festival*" which was previously stated, and the promotion of local artistry. These are unique assets and identities of Tuy that are worth recognizing by everyone.

To further support tourism development the LGU will also work on (1) Tourism site development, particularly on the provision of basic tourism facilities and services such as comfort rooms, local restaurants and/or food providers, souvenir shops, rest and first aid areas, information and directional signages, and adequate parking spaces; (2) Making tourism information easily available by having tourism information centers and local tourism office with the appointment of permanent municipal tourism officer and support staff; and (3) Identification of local cultural properties and conduct of cultural mapping for the preservation of Tuy's history and cultural heritage.

Since local income is largely IRA-dependent and local economic dynamism needs to catch up. Increasing the local sources of income is of vital consideration in the development process. In 2022, collection from business and trading is recorded at 4.3 million only. This is a very modest collection compared with other municipalities. **Commercialization** or the process of bringing in new businesses within the locality is a good option to increase local income. The municipality would also like to take advantage of the presence of the National Grid Corporation of the Philippines (NGCP), which has already unlocked opportunities for the energy sector. Support development thrust will likewise compose **industrialization**. However, industrialization in Tuy will be limited to power generation using alternative sources.

As a general guide to advancing the economic progress in the locality, **agriculture development** focus is the primary development thrust of Tuy. It will be supported by **tourism development**, **commercialization,** and **industrialization**.

III. LAND USE PLAN

A. Quick Review of CLUP-ZO Implementation (2000-2020)

At the time of 2000-2020 CLUP preparation, development concentrated in Poblacion, just like any other municipality in the Region. Palico Junction was perceived to be the alternative growth area where Tuy was successful in directing the development. Back then, properties along the road network were the practical choice to locate urban development. As expected, easy access and convenient mobility trigger the development, hence, linear development following road systems prevails today. Agri-industrial development had already emerged at that time. It was spurred by one of the well-known private companies, the San Miguel Corporation. It evolved into one of the major economic activities in the locality generating local income and employment.

On the other hand, Tuy failed to push for three significant proposals. These include: (1) The proposed and zoned ecotourism area in the eastern tip of the municipality never took off due to lack of support for infrastructure development. Bridge providing the direct link to Nasugbu requires a huge investment that is beyond the capacity of the local government unit; (2) The proposed and zoned quarry site in upland east specifically barangay San Jose, likewise, never utilized for such purpose. Instead, it was rehabilitated to create a forested area; and (3) The industrial zone located at the north western part of the municipality in Barangay Luntal were restricted by the land price hike.

It was noted from these observations that development drivers in Tuy include accessibility and convenience in mobility, government investment in infrastructure development, cost of land, and private promotion and investment.

B. Existing Land Use

Traditional land uses are common in the layback town of Tuy. Residential, commercial, institutional, cemetery, and agri-industrial dominate the land urban uses with an aggregated total area of 792.50 hectares. **Commercial use** is considered neighborhood type which supports the daily needs of the local community with few other establishments welcoming visitors like resorts and fast food. The absence of commercial banks in Tuy suggests low investment opportunities. The public market is the center of local commerce and trade for wholesale and retail. **Agri-industrial** facilities are focused on poultry production with few engaging in piggery. There are 23 poultry farms distributed in eight barangays namely Bayudbud, Guinhawa, Toong, Palincaro, Sabang, Talon, Bolobok and Magahis. It has been decided by the LGU that for the time being, additional poultry farms will not be accepted. Organic and planned **residential areas** are observed in both urban and rural barangays. While row houses exist, the majority of residential subdivisions are considered high-end. Bungalow-type residences are most common with few duplexes and multi-level designs. Few Spanish design ancestral homes still exist in barangays Rizal, Luna and Burgos. With an average lot size/household computed at 228.69 square meters residential density is considered moderate. **Institutional land uses** are composed of government offices, places of worship, educational facilities, and health facilities for community use. Burial grounds located in Barangays Mataywanak, Rizal, and Luna are still sufficient at the moment.

Roads occupied approximately 116.80 hectares providing inter linkages among different barangays. Road widths for both urban and rural areas are considered narrow with an average- width of 3 meters and 5 meters respectively.

Agricultural use is predominantly utilized for the production of sugarcane. However, the closure of Central Azucarera Don Pedro, Inc. (CADPI) in Nasugbu, Batangas triggered more crop diversification in some areas.

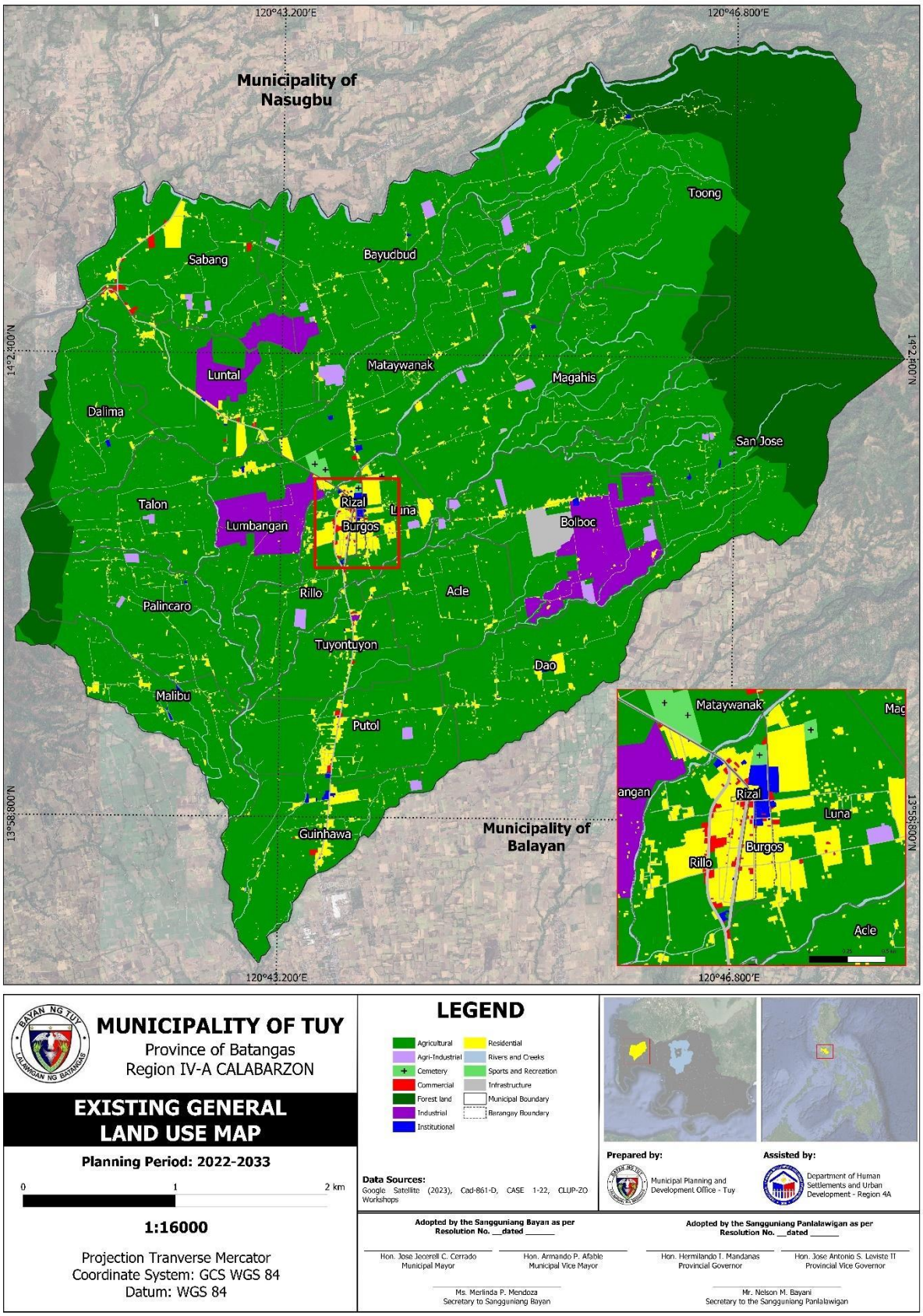
Approximately 15.96 percent of the total municipal territory is considered **Forested areas**. A portion of which is under DENR's rehabilitation Program and also considered for a local watershed. The rest of the forested areas found in the northeast of Tuy is a potential ecotourism destination. In fact, the LGU started to develop bike loops that were used prior to Pandemic. Locked down policies and health protocol restrictions affected the operation and retrieval became less priority. **Natural waterways** in Tuy have a total area of 176.63 hectares with preserved legal easements.

Table 2. Inventory of Existing Land Uses, 2022

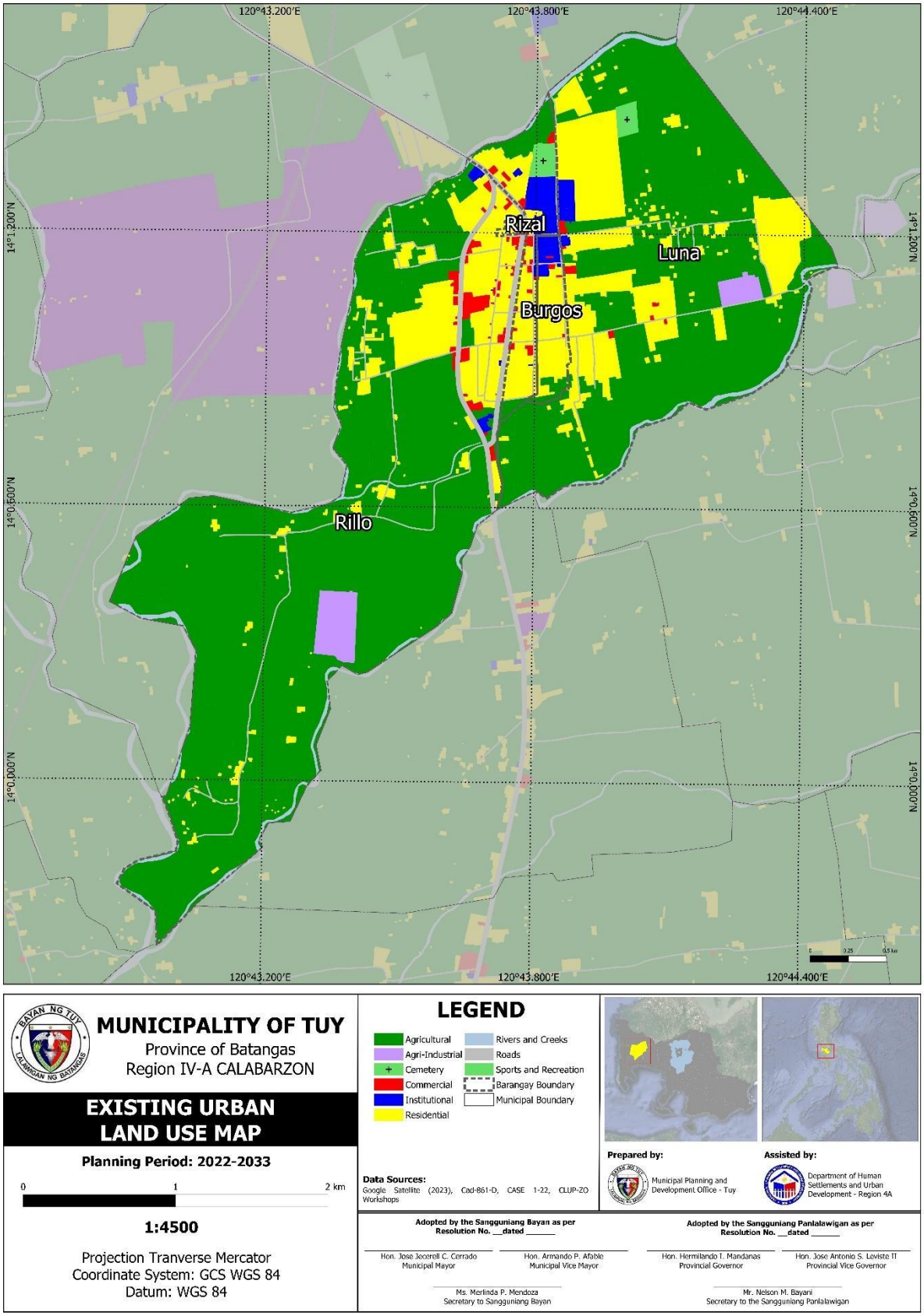
Land Use	Area (in hectares)	Percentage
Agricultural	6,854.80	71.79
Agri-Industrial	65.98	0.69
Cemetery	10.95	0.11
Commercial	14.87	0.16
Industrial	92.35	0.97
Specialized Industrial	293.18	3.07
Institutional	16.53	0.17
Parks and Recreation	0.15	0.00

Forest	1,523.48	15.96
Residential	298.49	3.13
Rivers and Creeks	176.63	1.85
Roads	116.80	1.22
Solid Waste Management Facility	0.53	0.01
Others	83.17	0.87
Total	9,547.92	100.00

Source: Google satellite combined with ground validation



Map 1. Existing General Land Use Map



Map 2. Existing General Land Use Map

C. Structure Plan

To guide the allocation of the proposed land uses, the structure plan identified the role of each barangay. The general hierarchy of barangays is likewise illustrated in this section for future development. This will serve as a spatial framework that typically shows the development scale or the change in the spatial layout within the territorial jurisdiction of the municipality. Using the historical trend analysis approach, the four policy areas namely production, protection, settlement growth areas, and infrastructure were identified and hereunder presented.

Primary Growth Area (PGA). Poblacion, composed of four (4) barangays: Burgos, Rizal, Rillio, and Luna, is the primary growth center serving the entire populace in terms of commerce and trade, government services, and social services. As the business and service center, it maintains external and internal connections both physically and economically.

Secondary Growth Area (SGA). The transformation of two barangays (Luntal and Sabang) into secondary growth areas is becoming evident. The role of the SGA is to link the interior rural barangays with basic services. In the future, barangay Luntal will overtake Poblacion's role of providing external-internal connectivity. The development of the grand terminal is also expected to create another business center with a different character. Unlike Poblacion, businesses that will preferably be located in SGA are more of a services type rather than the traditional buying and selling of basic goods. Likewise, Sabang as another SGA, will have significant development in commercialization.

Growth Areas (RGA). Out of 22 barangays, four (4) barangays namely Mataywanak, Tuyon-tuyon, Putol, and Bolbok will develop into growth areas (RGA). Most of these barangays will be providing community needs daily and some basic services similar to PGA and SGA but on a neighborhood-type scale only. Among the growth areas, Mataywanak is expected to advance as the institutional center as the prospective site for the new municipal hall, tertiary education facility, and evacuation center. Other growth areas will be triggered by the formation of residential subdivisions, government projects, and private investment.

Production Areas. Sugarcane production is found in every barangay though selected barangay produces specific crops that are unique in their respective area. Sabang, Luntal, Mataywanak, Toong, and Putol are now producing dragon fruit. Palincaro, Dalima, Talon, Guinhawa, and Bulboc are engaged in livestock and poultry production. Other barangays are producing different kinds of vegetables including coffee, banana, cassava, and yellow corn.

Protected Areas. Despite having a unique landscape. Mt. Batulao is not subjected to the Network of Integrated Protected Areas (NIPAS) Law. However, Tuy opted to protect Mt. Batulao which covers Toong, San Jose, and Dalima. In a study conducted by Londonio et.al 2020, it was stated that Mt. Batulao is a primary grassland ecosystem with different species of grass and insects to be found although there is a secondary forest established in steep slopes (DENR,2018). The area has been open for ecotourism for many years, which the LGU intends to maintain. Among other areas that deserve to be protected are the forested areas that contribute to good water percolation, hence, allowing replenishment of the groundwater supply. Groundwater sources in Luntal, Putol, and Poblacion areas will be part of the protected areas.

Places of Interest for Tourism Development. Among the places of interest that are worthy of tourism development include: (1) the largest river channel traversing Malibu; (2) the paragliding site and the agricultural areas of Luna; (3) the trail hiking site near Mt. Batulao in Toong; (4) bike trail going to Mt. Batulao in San Jose; and (5) the Lagunlong Falls in Rillio.

Infrastructure Support to Link Development Areas. Physical connectivity plays a vital role in driving local development. Fortunately, nationally funded major road systems are awaiting implementation to benefit Tuy. The Nasugbu-Bauan Expressway will traverse Putol and Dao while the 30-meter Tamayo Calaca City-Tagaytay-Lemery Diversion Road will pass through the existing barangays roads improving travel convenience. The locally initiated 6-meter circumferential road is also a major improvement in the local road systems.

D. Demand Supply and Analysis

Additional Area Requirement

The basic land uses for future urban requirements were estimated using the existing standard stipulated in the HLURB enhanced CLUP guidebook Volume 2. In the absence of necessary information to facilitate the computation, an alternative method was used like the proportion ratio and the requirements based on the proposed programs and projects already identified by the LGU. Additional households of 1,642 and 171 households that are due for relocation and an average household size of 4.1 were used to estimate the additional area requirement for residential use by the end of the planning period.

The computation shown in the table below was undertaken to determine the minimum requirement for reclassification using the projected population. Other assumptions were presented in detailing the proposed land uses.

Table 3. Demand Analysis for 2022-2033

Land Use Category	Standard	Estimated Additional Area Requirement by 2033 (ha.)
Residential	64m^2/HH (SH) 228.69 m^2/HH (based on average res. lot)	1.09 37.55
Commercial	based on existing commercial area to population ratio (13.42:48,357)	1.85
Light industrial	0.008/ha/person (considering 55,024 population)	44.01
Parks and recreation	500m^2/1000 population	2.75
Rural Roads	1.5km/100 ha.	
Burial Ground	based on 4.5 CDR and 127% PR	0.0767
Sanitary Landfill	based on .0211 kg/capita/day waste generation	1.927

Areas Suitable for Future Urban Development

It is imperative to identify the areas that are most suitable for urban development to lessen the development cost and government spending on regulation and monitoring during plan implementation. Based on existing laws and the physical characteristics of Tuy, urban development should not be located in the following:

Areas that are susceptible to flood and landslide. Unlike other municipalities, hazard susceptibility is not much of an issue in Tuy. Only 90.24 hectares are highly susceptible to flooding and 101.25 hectares are moderately susceptible. With regards to landslides, 1,687.96 hectares is highly susceptible, and 1,231.98 hectares is moderately susceptible. Combined flooding and landslide 1,778.20 hectares are deducted from the total land area, considering high susceptibility to hazards, which should not be utilized for urban development.

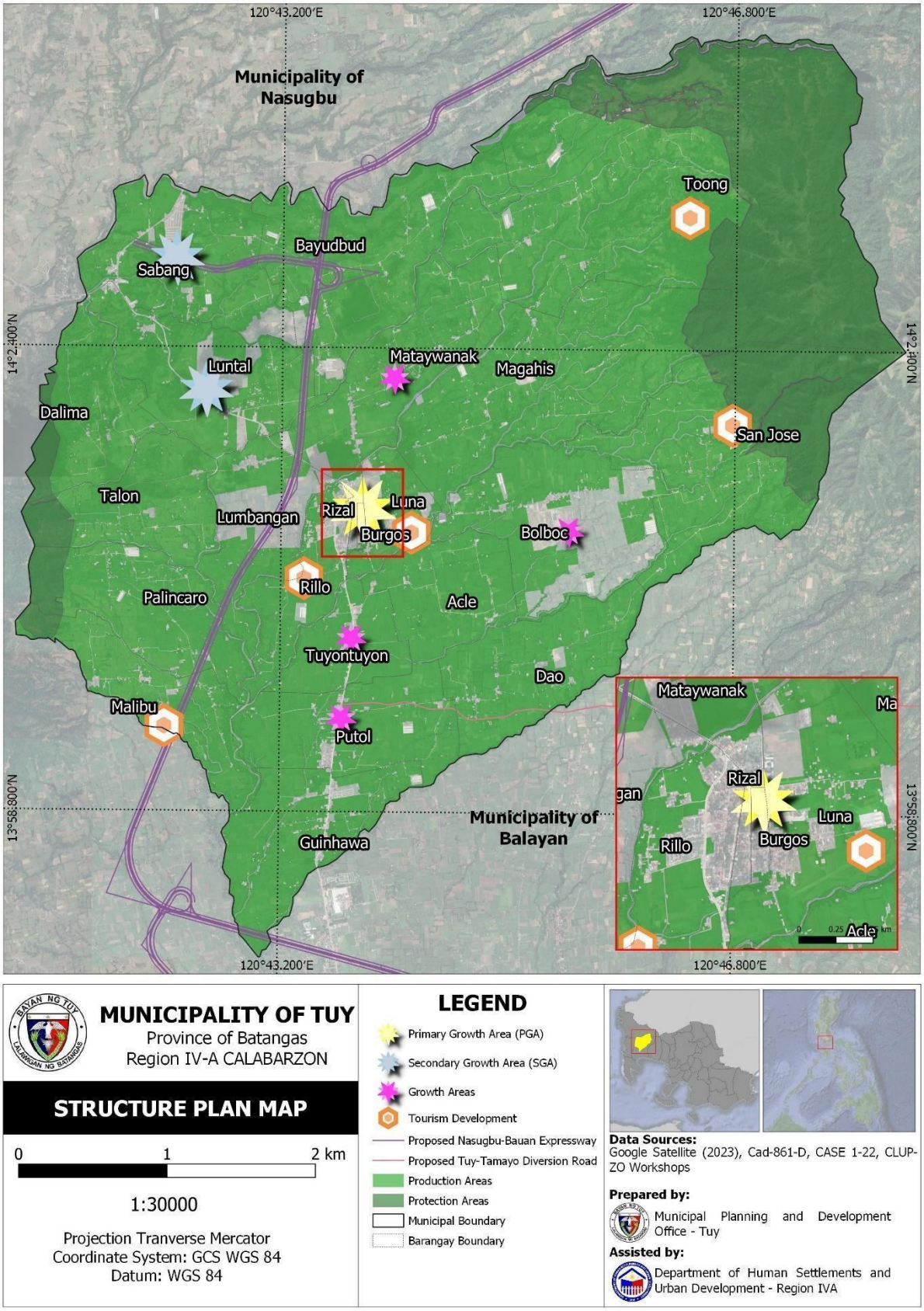
Waterways. The 176.63 hectares of rivers and creeks that serve as a natural water drainage shall be free from any form of obstruction and shall maintain a legal easement of 3 meters if traversing urban areas, 20 meters if traversing agricultural areas, and 40 meters if traversing forestland. [Philippine Water Code]

Forest land. The total land area of 1, 523.48 with a legal land status of forestland shall remain as forestland for purposes of balanced ecology. Forested areas must be maintained to effectively function for carbon sequestration, natural water cisterns, protection of biodiversity, and many others. (Refer to PD 705 and other related laws)

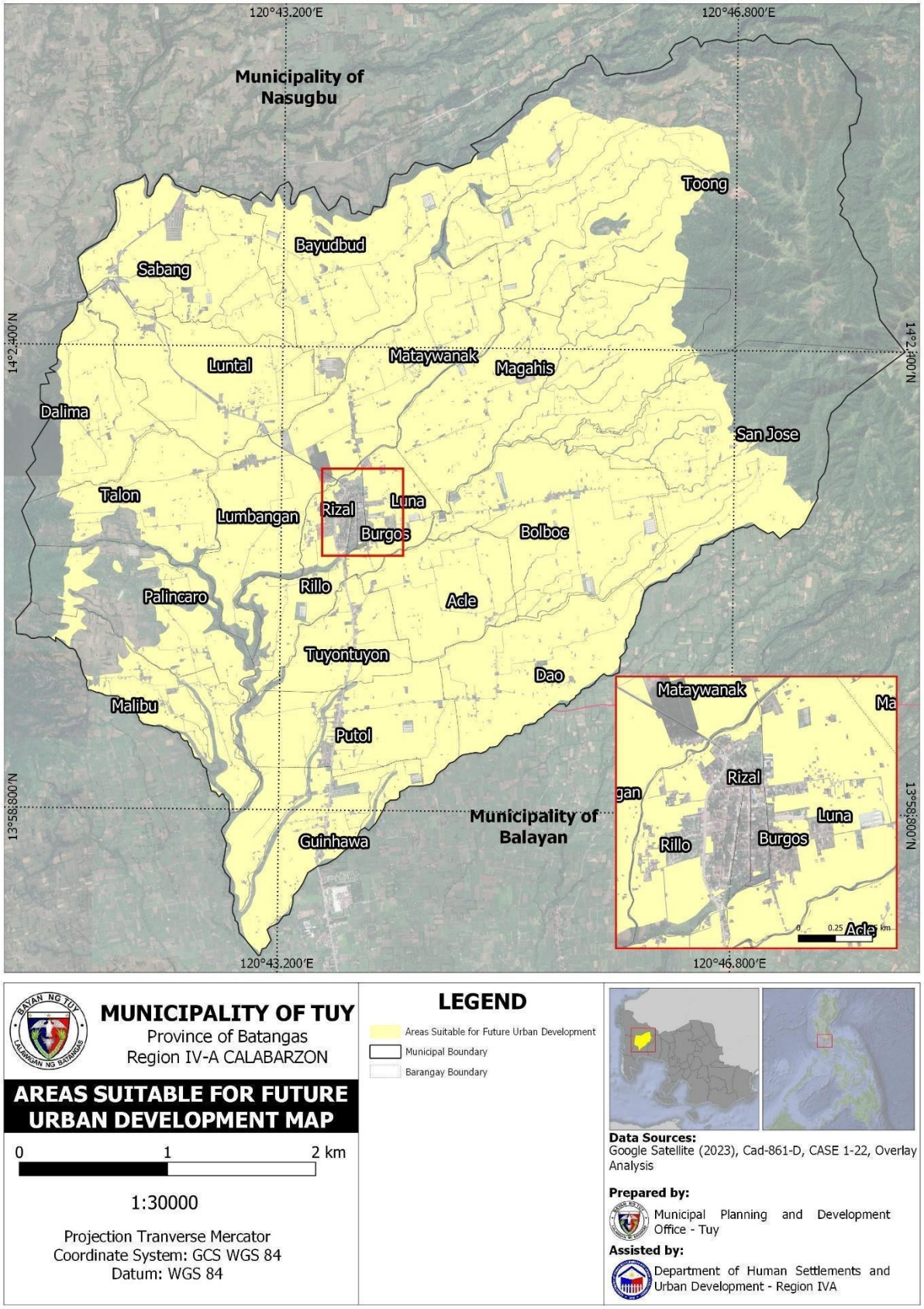
Table 4. Estimated Areas Suitable for Future Urban Development

Barangay	Areas Suitable for Future Urban Development (ha.)	Barangay	Areas Suitable for Future Urban Development (ha.)
Guinhawa	302.73	Mataywanak	381.57
San Jose	427.26	Burgos	0.24
Luntal	398.97	Lumbangan	258.08
Toong	709.09	Rillio	143.08
Tuyontuyon	177.30	Sabang	401.44
Dao	361.03	Acle	259.86
Bayudbud	482.71	Luna	112.77
Malibu	183.85	Talon	231.27
Putol	329.08	Magahis	670.14
Palincaro	170.49	Dalima	235.53
Bolboc	482.36	**TOTAL**	**6,721.26**
Rizal	2.41		

Source: Spatial Overlay Analysis using QGIS (2023)



Map 3. Structure Plan Map



Map 4. Areas Suitable for Future Urban Development Map

E. Detailing the Proposed Land Uses

In the conduct of determining the minimum land requirement for future urban development, population projection is the primary consideration. However, to have a good foresight of what is likely to transpire within ten years, impacts of external forces should be anticipated. While Tuy adjoins developing municipalities, it remains laid back but is slowly gaining momentum to take off. The land use proposal of the municipality intends to focus on effectively responding to the following major developments that are taking place, considering their potential impacts in Tuy, and the guiding principle of achieving inclusive growth and resilience.

1. Real Estate investments are on the rise increasing in-migration in the coming years;
2. Major transport developments such as the Cavite-Tagaytay-Batangas (CTBEX) expressway project and Nasugbu-Bauan Expressway (NBEx), will significantly enhance the accessibility of the municipality. This will open great investment opportunities to boost the local economy;
3. The closing of Centro Azucarera Don Pedro Inc. (CADPI) will encourage the diversification of farm products and engagement in agritourism activities to add value to the agriculture sector. It will expand the economic structure providing more local employment;
4. The eruption of Taal Volcano uncovered Tuy as a safe municipality relative to volcanic hazards, including floods and landslides. It is now the top choice for place of residents by extremely affected municipalities at a proximate distance;
5. The development of the National Grid Corporation of the Philippines (NGCP) sub-station draws major investment in renewable energies along with other opportunities; and
6. The development of a new provincial hospital enhances the competitiveness and relevance of Tuy as a municipality.

Urban Land Uses

Since 2000, residential development has been influenced by the road development system. People tend to locate where they can easily connect to goods and services and utilize public transport for moving from one place to another. Because it is easy to direct development along the roads for practical reasons, residential-commercial use will remain along the roads with the dominance of commercial use along major thoroughfares. Selected interior roads will be dominated by residential use. Hence, formation of commercial-residential corridors will be more prominent. Specifically, the following are proposed for **residential-commercial corridor:**

- Dominated by Commercial Use mixed with residential and institutional
  - 75 meters along both sides of the National Road passing Luntal and Sabang, Tuyon-tuyon, Guinhawa, Putol
  - 50 meters on both sides of Calaca City-Tagaytay-Lemery Diversion Road

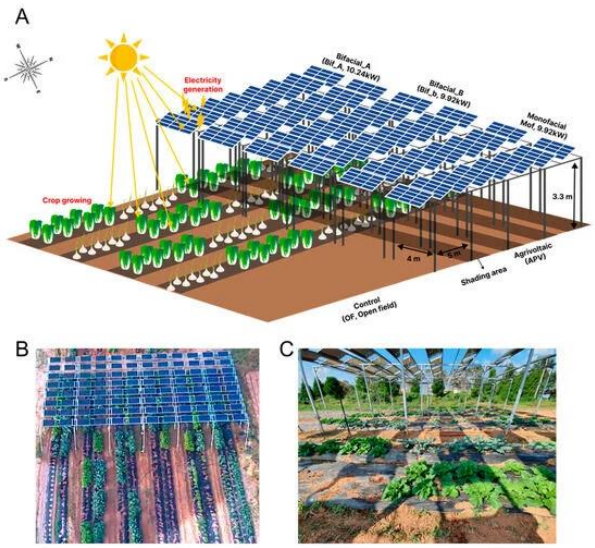
- Dominated by residential mixed with neighborhood commercial and institutional within the 50 meters of both sides
  - Luna-Bolboc-San Jose Barangay Road
  - Built-up area on the southwest of Mataywanak (Bayudbod-Mataywanak-Poblacion Barangay Road)
  - Lumbangan to Talon Barangay Road

In the Poblacion area, vacant areas within the boundary of Poblacion barangays shall be allotted for **residential expansion**. Upgrading of land uses at the core of the Poblacion will be adopted, transforming residential into commercial. The same will be applied in Sabang-Luntal intersection.

The southwest of Mataywanak adjacent to Luntal and Lumbangan will be allotted for **mixed commercial-institutional area with the dominance of institutional use**. This is in lieu of the structure plan designating Mataywanak the prospect of an institutional center. Functional parks and open space shall also be allotted within the institutional center.

Considering the existence of commercial warehouses in Tuy, and the Tuy Spur tool in the CTBEX alignment, **warehousing business** is anticipated to flourish in the locality. The proposed areas for warehousing businesses should not be more than 1km from the Tuy CTBEX entry/exit - Tuy Spur Tool, to minimize the number of huge delivery trucks entering and passing Tuy's major road systems.

Pockets of built-up areas which are dominantly residential distributed in different Sitios will be maintained with infilling. Scattered structures of residential, commercial and institutional that are not yet considered built-up areas will be maintained but shall be integrated as part of agricultural zones in the zoning ordinance.



Industrial areas that are currently occupied by the NGCP substation in barangay Bolboc, and solar farm facilities shall be designated as industrial areas. Tuy will welcome the development of solar farms and other facilities devoted to generation of power using alternative sources. However, the design should not restrict production of agricultural areas. Since solar farms demand use of huge agricultural areas, **agrivoltaics** should be adopted. Agrivoltaics allows solar energy production to co-exist with agriculture. As planned, solar farms are expected to be located in barangays Bolboc, Lumbangan, Luntal, Sabang, Magahis and Toong. An increase of **493.48 hectares** are allocated for industrial uses.

On the other hand, **agri-industrial areas** devoted to poultry and piggery shall be, likewise, maintained subject to regular or spot monitoring by the local government unit. Development of additional poultry farms, shall be subject to the local zoning ordinance formerly passed. Until revoked, the number of poultry farms shall not exceed 23.

Other Essential Urban Land Uses

An area for socialized housing is proposed in barangay Mataywanak with an area of **1.20 hectares.** There are still some housing units that are located along danger zones/areas such as riverbanks, creeks, and mountainous terrain which are susceptible to hazards such as landslide and flooding. Based on 2020 CDRA data, a total of 171 households are either made from light, semi-concrete and concrete materials.

Moreover, based on 0.0211 kg/capita/day waste generation, at least 1.9 hectares should be set aside for **sanitary landfill**. Tuy is already utilizing three (3) hectares for a Municipal Waste Management

Facility that include Sanitary Landfill Category I, located in barangay Magahis. The facility also serves as MRF with necessary equipment like composter and pulverizer. It is more relevant to consider site development since there is ample space for such purposes.

Based on demand estimation, less than one (1) hectare is needed for a **graveyard/public cemetery** or a place to honor the deceased. As part of Filipino culture and tradition, we honor and value our deceased loved ones by creating suitable and decent places for their remains. Since public graveyards should remain sufficient even way beyond the planning period, the municipality proposed an additional **0.81 hectares** of expansion of the cemetery in barangay Mataywanak.

The legal easements along river systems shall be established as **protection greenbelts** which shall be in consonance with the Philippine Water Code. Protection greenbelts shall be 3 meters if the river channel is crossing urban areas, 20 meters of agricultural areas, and 40 meters if within the forestland. The protection greenbelt will also serve as a linear **green park and open spaces** which should be the priority areas for planting of bamboo to prevent erosion and development encroachment. In some areas where extreme river erosion is experienced, support infrastructure is installed. If erosion is still manageable, planting of bamboo is more preferred. Tuy intends to sustain this practice. Unlike other LGUs, converting the legal easements in Tuy into green parks and open spaces are still convenient since encroachment is still minor to none. This planning period is just timely to protect the legal easement from development encroachment. One of the most valuable surface water resources of the Tuy is Malaking Ilog. To encourage community support in keeping the Malaking Ilog in its desirable state, it shall be opened to ecotourism activities. Though the primary purpose of greenbelts is more of improving the environment, the same may be also considered in developing a tourism portfolio. An area of **232.12 hectares** are allocated for a protection buffer in the municipality.

Forest and other Forest Use Category Land Uses

The existing land use would show that the total land area of 1,523.48 hectares is forestland but not 100 percent is forested. Despite having unforested areas, the entire forestland should remain forest for the purpose of balanced ecosystems. More often, forestlands which are accessible to the community are easily degraded due to maximizing the economic benefit of the forest rather than protecting its natural state for long term benefit. It has always been a challenge in land use planning to propose for best use within the forest in such a way that the community will give more value to long term benefit rather than just looking at the economic aspect. Just like any other municipalities with the same experience, ecotourism seemed to be the best option for dealing with these circumstances.

Tuy is adopting this concept in managing the forestlands. However, it is crucial to emphasize that while ecotourism is expected to spur economic benefit of forest resources, the development intention is to give utmost consideration to its value as a natural resource. In this regard, critical forest areas that are considered natural water cistern shall be designated protection forest. This has a total land area of **557.34 hectares**. The remaining forestlands outside of the protection areas shall be considered production forest where agroforestry shall be applied which accounts to **734.93 hectares**. Areas that have been rehabilitated by the DENR through its reforestation program shall be maintained subject to the rules and regulations stipulated in the stewardship contracts, if any.

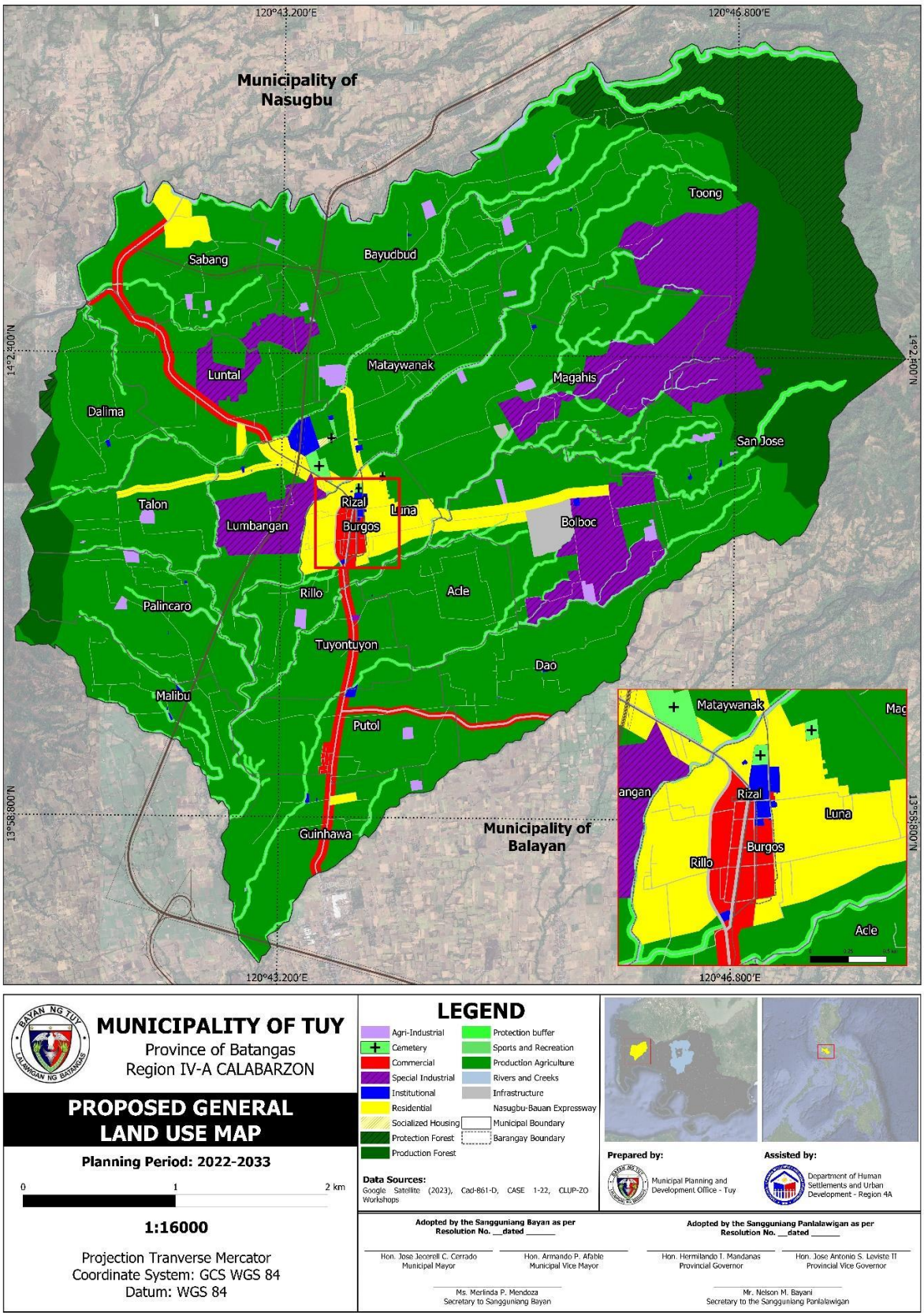
Agricultural Land Uses

Because agricultural areas are open to all types of urban development, it is most vulnerable to reclassification and subsequent conversion. Especially today, the sugarcane industry is dwindling as mentioned in development constraints. The best approach to managing agricultural lands is to inculcate among the local farmers and rural residents the value and benefits of ecology, their greater eco-sensitivity and the greater attentiveness to the environment and nature protection. Agritourism or farm tourism is one of the few non-agricultural activities which triggers the development of agriculture by creating a favorable market for agricultural products and increases the farmer's personal income. Agritourism in the rural areas can also ensure additional jobs for the rural residents as, frequently, it substantially generates the development of tourism-related sectors, especially services, commerce, food serving, and processing. Tuy, therefore, will open agricultural areas to agritourism. However, agritourism shall not be designated as a separate land use, instead it will be treated as an integral component of the agriculture areas.

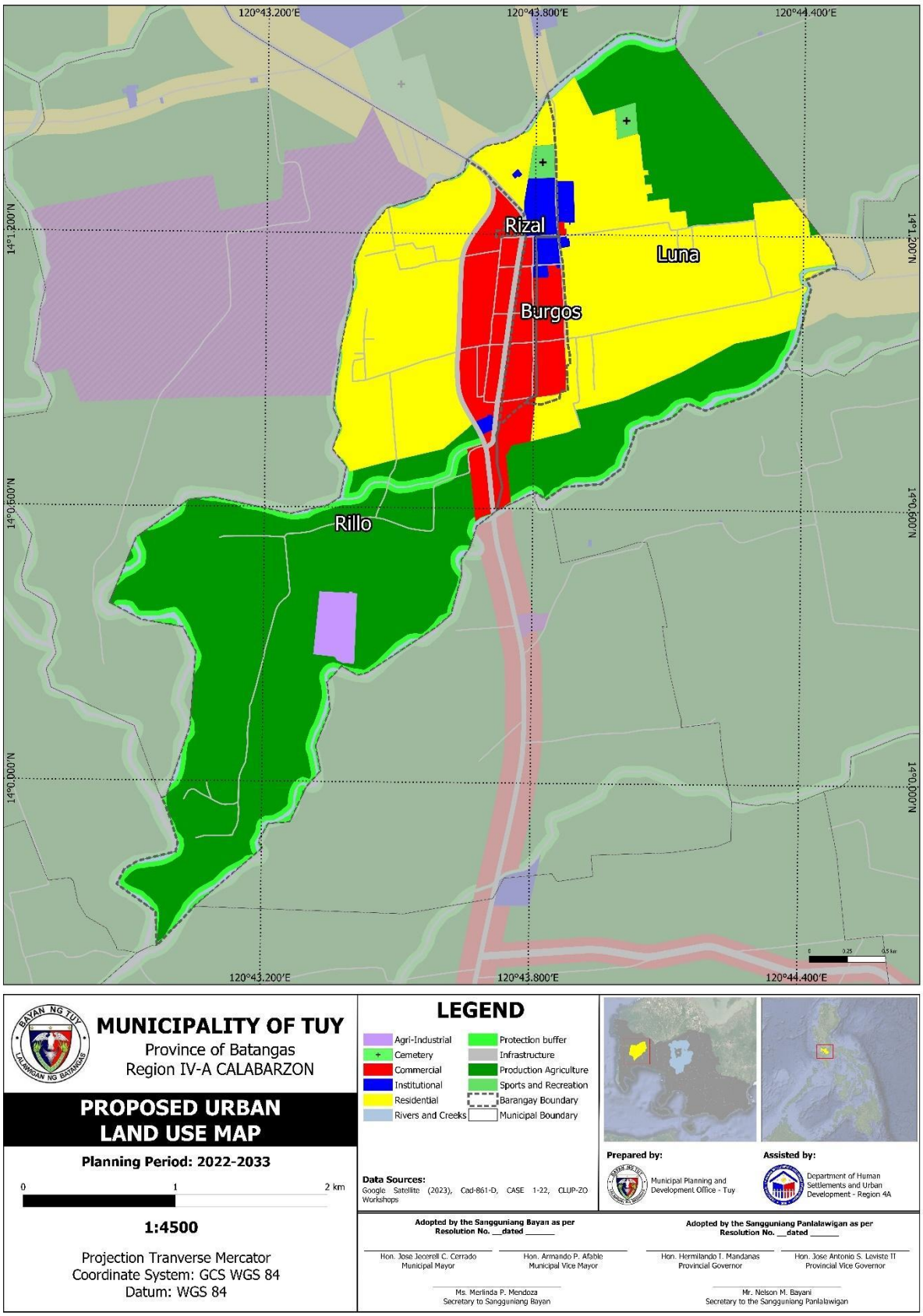
Other land uses that will be integrated within the agricultural areas are the dispersed residential uses that are used as home lots of local farmers. These commonly exist in more than 500 square meters surrounded by vast agricultural areas.

Quarrying

Quarrying has been operating in the municipality for several years particularly in barangay Toong. The LGU opted to keep quarrying in this planning period subject to land use policies to minimize the projected impacts based on the present operation.



Map 5. Proposed General Land Use Map



Map 6. Proposed Urban Land Use Map

F. Change in Land Use, 2022-2033

The table below presents the change in land use in the municipality of Tuy for the planning period 2022-2033. Initially, the existing forest area is further proposed into production and protection forest with an area of 734.93 hectares and 557.34 hectares, respectively. A total of 231.21 hectares were also allocated as a protection buffer within the forest area following the 40-meter buffer of the Water Code. Largest areas increase is proposed for industrial uses (879.01 hectares), residential uses (297.93 hectares), and commercial areas (166.24 hectares). The greatest increase in land use area is accounted for the commercial uses with an increase of 1017 percent. On the other hand, decrease in areas accounted for residential areas are due to the designation of protection buffers. Additional areas allocated for protection buffers account for 232.12 hectares.

By income category, the Municipality of Tuy is only allowed to reclassify 10 percent of its existing agricultural areas or 685.48 hectares to non-agricultural uses as per MC54. Based on the presented table, there is a decrease of 909.51 hectares. However, 231.12 hectares was allocated for a protection buffer. Thus, for the planning period of 2022-2033, the municipality of Tuy is only reclassifying a total area of 678.39 hectares which is only 9.89 percent of the existing agricultural areas.

Table 5. Change in Land Use 2023-2033

Land Use	Area (in hectares)			Percent Change
	Existing	Proposal	Increase/ Decrease	
Agricultural	6,854.80	5,945.29	-909.51	-13.3%
Agri-Industrial	65.98	68.18	2.20	3.3%
Cemetery	10.95	11.76	0.81	7.4%
Commercial	14.87	166.24	151.37	1017.7%
Specialized Industrial	350.90	842.17	491.17	140.0
Institutional	16.53	32.69	16.16	97.7%
Parks and Recreation	0.15	1.33	1.18	788.7%
Production Forest	1,523.48	734.93	-	-36.6%
Protection Forest		557.34		
Protection Buffer within Forest		231.21		
Residential	298.49	297.93	-0.56	-0.2%
Rivers and Creeks	176.63	176.64	0.00	-
Roads	116.80	125.80	9.00	7.7%
Solid Waste Management Facility	0.53	3.09	2.56	484.7%

Protection Buffer		232.12	232.12	-
Socialized Housing	-	1.20	1.20	-
Others	83.17	83.17	-	-
Total	**9,547.92**	**9,547.92**	-	-

Source: Google Satellite Image, CLUP-ZO Workshops, Ground Survey

G. Land Use Policies

Along with the land use proposal, Tuy tried to anticipate what lies ahead with the foresight of having linear development along the major road systems, the market forces, the private investments, the community needs and other development factors. With an end goal of achieving sustainable and inclusive growth, policy interventions were crafted to manage if not prevented the potential development adversity. There might be some interventions that are more suited to lessen development adversity, but at the time of preparation of this plan, Tuy managed to come up with the following policy intervention.

Table 6. Proposed Land Use Policies

Areas of Concern	Proposed Policy Intervention
Agri Industrial Development: The number of poultry farms is rising fast within the area. Carrying capacity might become a major concern in the future. While it is one of the local sources of income, the municipality should maintain only a good number of poultry to avoid environmental and health related issues. On the contrary, the multi million investment of some poultry farms are already at risk due to emerging investment on residential subdivisions. The poultry owners should have the same protection as well. The municipality is responsible to protect the interest of both parties.	H. Strict implementation of Municipal Ordinance No. 2023-02, regulating the number of poultry facilities that should exist within the municipality I. Residential subdivisions must not be located within a radial distance of at least 1 km to any existing poultry J. Existing poultry near the residential vicinity shall provide ample buffer planted with trees K. Poultry owner should implement corporate social responsibility (CSR) within the municipality L. Regular and spot monitoring team shall be created to ensure that poultry and piggery operation do not cause adverse effects in the environment specially in the water sources (ground and surface water sources) M. Agri Industrial operators must provide their own STP.

Areas of Concern	Proposed Policy Intervention
Agritourism Related Development: The concept of agritourism has been contributing to economic growth in many Regions. However, agricultural production should not be a secondary purpose. Tourism is an added value to agricultural development. Once it becomes more profitable, agricultural production may no longer be attractive. Agricultural areas applied with agritourism may gradually convert into purely tourism changing the character of the land.	● Integrated farming practices must be applied ● Agritourism development must be in consonance with the standard to be set by the municipal agricultural office ● Partnership with the existing local cooperatives should be given a priority in agritourism development ● Support structural development should be limited to only 20 percent of the entire area ● Integrate organic farming system ● The tourism development shall be in accordance with Republic Act No. 10816 or the Farmourism Development Act of 2016
Ecotourism Development: Development of support infrastructure for ecotourism activities may threaten the natural environment. Bike trails may damage forested areas and soil erosion may be experienced. Other necessary structures may be preferred over forest conservation policies.	● Support legislation on the management and regulation of ecotourism development
Waterways Protection: River systems within Tuy may not be at their best state today, but are still relatively in condition and suitable for water contact activities such as swimming. However, if protection is neglected while development is on the rise, contamination may reduce water quality. The water contact recreation activities will be restricted by poor quality.	● Establishing legal easement and converting the same into green parks and open spaces ● Any form of contaminants shall not be located within 50 meters from the riverbanks ● Ensure that farm lot developments do not pollute waterways through regular monitoring ● Ensure that poultry and piggery do not discharge waste to river systems
Conversion of Sugarcane areas into farm lot subdivision: With closure of major Azucarera in Batangas, the profitability of sugarcane production will significantly decrease, hence, may lead to conversion of sugarcane plantation into non agricultural purposes. In fact, it is now being manifested in some areas.	● No conversion of sugarcane areas that is considered of prime production ● Priority support to sugarcane farmers that are interested to venture into other products like corn, cassava and other crops

Areas of Concern	Proposed Policy Intervention
	● Provide ample technical assistance to sugarcane farmers to practice crop diversification
Water Resources Utilization. The growing demand for water supply is likely to create pressure on the water sector to satisfy the demand among competing users. This will be further aggravated by the impacts of climate change.	● Review policy pertaining to sharing water supply with the Municipality of Nasugbu ● Restrict extraction of groundwater supply for swimming pool ● Regular monitoring of private resorts' water sources
Continuous Quarrying. Development of adjacent municipalities is expected to cause a steady demand in quarry aggregates. If not properly regulated this may cause serious soil erosion and landslides and will continuously damage barangay access road in Barangay Toong	● Continuous collection of Municipal Share from Provincial Government for quarrying companies/ owners. Review existing policies pertaining to collection of fees from quarry operation. ● Determine the metes and bounds of the quarry site and establish clear boundaries ● Conduct of regular monitoring should
Utilization and Management of Forestland. While ecotourism will be adapted for economic gain, the LGU cannot discount the high probability of mismanagement within the forestland. The LGU's shortcomings to regularly and effectively monitor the activities within the forestland may not be easily addressed. It is important to engage the community in managing forest resources.	● Implement community-based forest management programs and projects in coordination with the DENR. ● Rehabilitation of the declared Watershed Areas and strict implementation of protection policies within the area.

H. **Other policies for specific land uses**

Similar with other LGUs within the province, the following existing policies for location standards will be used to guide specific developments hereunder stipulated.

Location Standards for Multi-modal Transport Terminal. The location of the transport terminal shall be subject to the following minimum site criteria pursuant to DILG and DOTC Joint Memorandum Circular No. 01 s. 2008:

1. More than 100 meters away from institutional establishments particularly schools and hospitals
2. The terminal must be accessible to commuters, i.e. transfer routes are available or within its service radius. However, direct access to major thoroughfares especially high speed

highways and expressways should be discouraged for safety and smooth traffic flow purposes The location of bus stations/terminals, freight/truck terminals should be outside the center of commercial activities to reduce and minimize street congestion

3. Jeepney/taxi terminal may be located within the central business district or commercial zone provided, it should not be major road intersection
4. Garage must be located at the outskirt of the business center or commercial zone

Location Standards for Memorial Parks and Public Cemetery. The location of the memorial parks and public cemetery shall be subject to the following minimum site criteria pursuant to HLURB Implementing Rules and Regulations for the development of the same:

1. Agricultural zone but not covered by CARP, unirrigated and should not be within the SAFDZ area;
2. Must be a 100 meters distance from residential, commercial, industrial or institutional;
3. Should be located outside environmentally critical areas as defined in Proclamation No. 2146;
4. Should be located on ground where the water table is not higher than 4.5 meters below the ground surface
5. Served by a road-right-way with a minimum width of 8.00 meters

Recreational and Tourism oriented development. All types of development that are considered recreational and tourism oriented in nature like resorts should promote water percolation at an acceptable level. For such a purpose, 30 percent of the total development area must remain permeable and/or unpaved. The same shall be compliant with the National Green Building Code Section 10-15, whenever applicable. To wit:

1. Energy Efficiency
2. Water Efficiency
3. Material Sustainability
4. Solid Waste Management
5. Site Sustainability
6. Indoor Environmental Quality

In anticipation of climate change impact on local water sources, and to uphold water efficiency, recreational and tourism oriented development shall be required to have their own water cistern. This is to encourage the collection and use of rain waters for business utilization.

IV. Implementation and Monitoring Scheme

The Comprehensive Land Use Plan (CLUP) of the municipality has an accompanying Zoning Ordinance (ZO) which includes the scheme for implementation and monitoring of the CLUP-ZO. This includes the composition of all special committees for the implementation and their corresponding duties and responsibilities. Overall, the Zoning Officer shall oversee these responsibilities together with the Local Zoning Board of Appeals (LZBA). Moreover, the Local Zoning Review Committee shall also be organized and work on its main function of reviewing the CLUP-ZO every three years. Lastly, proper coordination should be consistent with national agencies such as DHSUD, DENR-PAMO, Bureau of Fisheries and Aquatic Resources, Mines and Geosciences Bureau (MGB), PHIVOLCS and other related agencies. Volume 1 Land Use Plan Tuy, Batangas The implementing and monitoring scheme of the CLUP may be

referred to as national and regional plans, as well as to local plans such as the Comprehensive Development Plan (CDP), Local Development Investment Plan (LDIP) and Executive-Legislative Agenda (ELA) for prioritization of future developments. Lastly, the Communication Plan which aims to disseminate the contents of the CLUP-ZO to the stakeholders is also presented in Table 7.

Table 7. CLUP-ZO Implementation and Monitoring Action Plan

Action to be undertaken	Instrument	Year	Responsible Office/Agency
Communicate the approved CLUP-ZO to the stakeholders	Communication Action Plan	Year 1	LGU Collective Effort
Creation of Committee on Housing and Land Use	SB Resolution	Year 1	SB
Activation of local bodies that are deemed necessary for the implementation of the CLUP-ZO like the LZBA and the LZRC with establish internal processes	Executive Order	After the publication (Y1 of plan implementation)	MPDO and Office of the LCE
Designate Enforcement Officer	Executive Order	Upon Effectivity	Office of the LCE
Capability Training Program for the Key implementers: i.e., Enforcement Training; Reclassification Procedures	Project Proposal	Year one of the plan implementation or as the need arises	MPDO, Budget Office, DHSUD and DILG, DENR
Formulation of Supplemental Plans:	Project Proposal with SB Resolution	Year 1-5	Office of the SB and Office of the LCE
Tourism Master Plan			MTO and DOT
Review Schedule of Fees, Fines and Penalties for Land Use Development Regulations for Adoption	SB Ordinance	Year 1	SB and MPDO
Create a simple land use development monitoring system (manual and/or automated system)	Project Proposal	Year 1-3	MPDO; DHSUD and other partners
Institutionalize Regular Inspection/Monitoring of Land use changes	Executive Order and/or SB Resolution	Year 1-3	MPDO and MEO; DHSUD
Improve the monitoring tools and systems	TWG	Year 4 to 2032	MPDO; DHSUD; other partners
Conduct CLUP-ZO implementation and monitoring reviews	SB Resolution	Every 3 years	MPDO; LZRC

Action to be undertaken	Instrument	Year	Responsible Office/Agency
Installation/Maintenance/Improvement of Hazard Markings in High-risk Areas	SB Resolution	Year 1-3	MDRRMO, DILG
Adopt and Standardize Forms and Requirements for application of Zoning Certifications and Locational Clearances including flowchart of integrated permits and licensing system	SB Resolution	Year 1-2	MPDO, DHSUD
Inventory of Zoned lot parcels to monitor changes in land uses	Lot parcels inventory Matrix	Year 1-3	Assessor's Office, MPDO
Regularly revisit the DRRM Plan/Preparedness and Mitigation Measure Plan/Contingency Plan and the like which shall establish the pick-up points, evacuation sites, redundant/escape routes, infrastructure mitigation and others and ensure that the same is still applicable	SB Resolution	Every three years or as the need arises	MDRRMO, SB

Source: CLUP-ZO Workshops

Table 8. Communication Plan

Goal	Objective	Target Audience	Key Content / CLUP Salient Points	Activity	Communication Channel / Tool	Duration (From Preparation to Conduct and Post-Evaluation)	Timeframe (Month and Year)	Estimated Budget	Responsible Bodies
To communicate the salient points of the CLUP-ZO (2024-2034) to relevant stakeholders in the LGU Tuy, Batangas	To provide content of the information materials for dissemination	Stakeholders	CLUP overview, LGU's approved CLUP background and summary	Lay-outing and document checking	Leaflets, brochures	1-2 months	Last Quarter of 2024	P5, 000	DHSUD R4A, MPDO, IT
	To produce materials re CLUP-ZO information dissemination	Stakeholders		Printing-out of communication materials (e.g., leaflets, brochures, tarpaulin); Production of soft-copy materials	Leaflets, brochures, tarpaulins, e-copies for social media and website posting	1-2 months	Last Quarter of 2024	P5, 000 (LGU)	MPDO, IT, Barangay Offices
	To reiterate the overview of the CLUP-ZO Formulation to the employees, department heads and SB members of the LGU-Tuy	Employees, Department heads and SB members of the LGU-Tuy	Rationale, CLUP 12-Step Process, Review and approval process, Policy, HLURB/DHSUD Guidebooks	Face-to-face orientation/information dissemination	Presentation slides, Leaflets	3 months preparation (1 day activity)	Last Quarter of 2024	P10, 000	DHSUD R4A ELUPDD
	To inform the LGU personnel about the content of the approved CLUP-ZO (2024-2034)	Employees, Department heads and SB members of the LGU-Tuy	CLUP-ZO content (eg., Brief Profile, Land Use Proposal, Zoning)	Face-to-face information dissemination and posting of physical materials to offices	Presentation slides, Leaflets, brochures, posters	3 months preparation (1 day activity)	Last Quarter of 2024	P10-15, 000	MPDO, IT, Office of the LCE
		Stakeholders		Social media, website posting and CLUP-ZO summarized report (2-3 pages)	E-copies of posts (in JPEG format), captioned posts, summarized CLUP-ZO report, video presentation (AVPs)	3 months preparation (1 pinned post in MPDO FB page)	Last Quarter of 2024	P5, 000	MPDO, IT

Goal	Objective	Target Audience	Key Content / CLUP Salient Points	Activity	Communication Channel / Tool	Duration (From Preparation to Conduct and Post-Evaluation)	Timeframe (Month and Year)	Estimated Budget	Responsible Bodies
	To inform the Barangay Officials and CSOs about the content of the approved CLUP-ZO (2024-2034)	Barangay Officials and Civil Society Organizations (CSO)/Non-Government Organizations (NGOs)		Face-to-face information dissemination and distribution of physical materials to offices	Presentation slides, Leaflets, brochures, posters	3 months (At least twice a year)	First or second sem of 2024-2025	P10, 000	MPDO, LCE
	To inform the Barangay Officials about the content of the approved CLUP-ZO (2024-2034)	Barangay Officials		Face-to-face distribution of CLUP-ZO summarized report, brochures, posters	CLUP-ZO Summarized Report, brochures, posters	2 weeks prepartion (1-2 days activity)	Last Quarter of 2024	P5, 000	MPDO, Barangay
	To ensure copies of CLUP-ZO documents and materials for the MPDO for public use	Public/Stakeholders, Department Heads, DHSUD, DILG, Mayor's Office, Sangguniang Bayan Office	CLUP-ZO documents (e.g., Volume I, II and III); brochures	Storage and filing of CLUP-ZO document; Material production	CLUP-ZO Documents; brochures	1 month	Last Quarter of 2024	P30, 000	MPDO, IT, Budget Offices
	To provide a copy of flow chart/process flow of land reclassification and development permit application for the public/stakeholders' information and guide	Public/Stakeholders	Land Reclassification and Development Permit Application Process	Printing-out of flow chart and process flow in the MPDO and Information Board	Tarpaulin and e-copy in the social media page/website	2 months	Last Quarter of 2024	P5, 000	MPDO, Engineering Office, BFP, Barangay Office
	To explain the Model Zoning Ordinance, Zoning Enforcement and Land Reclassification to the MPDO and Local Zoning	MPDO personnel, Local Zoning Board/Committee	Model Zoning Ordinance and LGU ZO	Training Workshop, Focused Group Discussion	Lecture, FGD, Presentation slides	3 months	Last Quarter of 2024	P50,000	DHSUD R4A-ELUPDD, MPDO, Department Heads

Goal	Objective	Target Audience	Key Content / CLUP Salient Points	Activity	Communication Channel / Tool	Duration (From Preparation to Conduct and Post-Evaluation)	Timeframe (Month and Year)	Estimated Budget	Responsible Bodies
	Board/Committee in LGU Tuy, Batangas								

Source: CLUP-ZO Consultation Meeting

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Agrivoltaic Project. Photo by Wa r.e. posted in https://balkangreenenergynews.com/